Eric Xie

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Education

University of Toronto (UofT) Toronto, ON

BASc. Engineering Science *Sep. 2024 – Apr. 2028*

- GPA: 3.80/4.00
- Extracurriculars: aUToronto (multimodal perception lead, creating auto-labelling pipeline to train BEVFusion lidar & camera detection backbone to enable Canada's first L4 self-driving car); UofT Aerospace SAE software team (camera simulation & localization in Gazebo); UofT Formula Racing controls team (implementing LTV-MPC)
- Roles: EngSci 1st & 2nd year class representative - resolve academic challenges, improve student well-being

Experience

Robotics Simulation SWE Intern July 2026 – Present

NVIDIA *Toronto, ON*

- Isaac Sim, in progress

Robotics Researcher May 2026 – Present

Autonomous Space Robotics Lab (ASRL) *Toronto, ON*

- First-authored a paper advised by Prof. Timothy D. Barfoot, currently under ICRA review.
- Developed an automatic lidar-to-radar labelling pipeline to generate temporally consistent labels across 643 km of the Boreas Road Trip dataset.
- Demonstrated that Doppler correction on spinning radar images improves mAP by up to 2.37 points across state-of-the-art detectors.
- Designed a novel Doppler-based velocity estimator for per-vehicle track initialization, improving MOTA by 13.68 points and achieving 99.7% of performance using ground-truth velocity priors.

Robotics Research Intern May 2025 – Sep. 2025

Syncere AI *Toronto, ON*

- Allowed robot arms to be implicitly compliant and reactive to external forces, improving safety in human-robot interactions while removing need for a 6-axis torque sensor which reduces cost by at least $2,000
- Designed an end-to-end impedance controller in Isaac Lab, trained with RSL-RL PPO library
- Overcame sim-to-real gap with domain randomization and noise injection, and deployed on Koch arm robot
- Developed a bilateral teleoperation system using servo current readings, boosting data collection speed by 17%
- Collected teleoperation data and trained with diffusion to allow for compliant movements during inference

MLH Software Engineering Fellow June 2024 – Sep. 2024

Facebook AI Research *Remote*

- Contributed to SimulEval, an evaluation toolkit for simultaneous translation; 5 Python pull requests merged
- Created visualization features for raw speech to text data, aiding researchers in understanding output
- Implemented a Docker feature for remote speech to text inference, ensuring uniform environment dependencies
- Developed a streaming demo feature that transcribes speech in real-time through OpenAI's Whisper and Pyaudio
- Proactively troubleshot logic and dependency errors, improving GitHub workflow efficiency by 50%

Projects

REACT | *sEMG, Transformers, CNN, GRU, FiLM Conditioning* Dec. 2025 – Jun. 2026

- Developed a conditioning framework for user-adaptive sEMG hand pose estimation to overcome cross-user generalization challenge; designed the calibration module with transformer encoders and FiLM conditioning
- Beat Meta's emg2pose public dataset benchmark; achieved consistent 1.6% to 3.9% error reductions across User, Stage, and User+Stage held-out groups for both regression and tracking tasks
- Presented research at two workshops at ICRA 2026 @ Vienna, the largest robotics conference in the world

DUM-E | *Robotics, NLP, OpenCV, Raspberry Pi, Groq, Llama 3.3* Sep. 2025

- Invented DUM-E from the Iron Man movie - an NLP powered robotic arm that can reason, generalize, and execute tasks such as grasping, waving, and doing pick & place - with just voice, winning Waterloo's Hack the North

- Analytically solved 5 DOF inverse kinematics; successfully deployed on Raspberry Pi with inference time of <1 ms
- Leveraged OpenCV and homography to locate objects; created multimodal VLA with Groq & Llama 3.3

Autotune | *TypeScript, Next.js, Python, Firebase, HuggingFace API* Mar. 2025

- Built a marketplace for anyone to outsource ML model fine-tuning, democratizing access to GPU computing resources, winning Best ML Hack at GenAI Genesis
- Created frontend with TypeScript; SSR with Next.js to optimize performance; integrated with Python backend to submit training requests in Firebase and finetune with HuggingFace API

Nail Braille | *Arduino, CAD, Mozilla DeepSpeech* July 2022 – Nov. 2022

- Chief Engineering Officer: Led development of a speech-to-braille learning device for visually impaired individuals to help navigating and interacting with the surrounding environment
- Utilized Arduino & CAD; interfaced Mozilla DeepSpeech; developed 3 iterative prototypes
- Shad 2022 Best Prototype Award & 2022 Youth Innovation Showcase Finalist + Public's Choice Award

VEX Robotics | *C++, PID, Odometry, CAD* Sep. 2021 – June 2024

- Programmer & Designer: Implemented motion control algorithms (PID, odometry, pure pursuit) to execute precise autonomous robotic movements
- Designed with CAD; constructed hardware prototypes and mechanisms: power-take-off (PTO), chassis, ratchet
- 8 tournament awards, including 2022 BC Provincial Finalist and scoring top 30% in the World Championship

Awards & Certifications

ICRA 4th HRI NeuroDesign Workshop Competition Finalist, 4th/11 | *Presented REACT* June 2026
ICRA Robotics Dextrous Manipulation Workshop Acceptance | *Poster and lighting talk* June 2026
Hack the North Winner | *NLP powered robotic arm* Sep. 2025
AWS AI & ML Scholarship | *End-to-end ML workflows in AWS with Sagemaker* Aug. 2025
ICRA Undergrad Outreach Workshop Award | *Attending world's largest robotics conference* May 2025
GenAI Genesis Hackathon Best ML Hack | *LLM finetuning marketplace* Mar. 2025
UofT Engineering Kompetition (UTEK) Junior Division Finalist Jan. 2025
AWS Hack The Student Life Hackathon Winner | *Roommate matching using semantic search* Nov. 2024
International Robotics Honour Society Jan. 2024
ACSL Senior Division 1st Place, Finalist | *Competitive programming* May 2023
USA Computing Olympiad (USACO) Gold Division | *Competitive programming* Mar. 2023

Technical Skills & Interests

Software Skills: C++, Java, Python, PyTorch, HTML, CSS, JavaScript, ROS, Docker, GitHub, ReactJS, NextJS, Vercel, Isaac Gym, Gazebo, OpenCV, AWS Sagemaker, S3, Lambda & Step functions, CUDA, Verilog, RISC V Assmebly
Hardware Skills: Solidworks, Arduino, Onshape, 3D Printing, Soldering, Machining, Welding, Robotics design
Professional Skills: Leadership & Management, Teamwork & Collaboration, Active learning & Curiosity, Communication & Public speaking
Languages: English (native), Mandarin (native), French (intermediate)
Hobbies: Track and field, ultimate frisbee, swimming; Piano, finger style guitar; Chess (Chinese Candidate Master)

Leadership

Tournament Manager Oct. 2021 – June 2024
Ten Ton Robotics *West Vancouver, BC*

- Reserved VEX IQ elementary & middle school tournament venue; set up hardware & software; coordinated volunteers; resolved logistic & technical challenges
- Ten Ton Robotics Most Valuable Volunteer 2022 & 2023
- International Robotics Honour Society 2024 for exceptional contribution in robotics education

VEX IQ Robotics Instructor July 2023 – Aug. 2023
UTG Academy *Vancouver, BC*

- Taught hardware mechanisms: bar lifts, gear ratios to educate young students about mechanical design
- Introduced software tools: Scratch, VR to help students build fundamental programming skills
- Fostered an interactive and fun play-and-learn environment for 20+ middle school VEX IQ students